

Cataract Surgery Phase Recognition Using Temporal Deep Learning Architectures

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Abstract—Automatic surgical phase recognition from cataract surgery videos can support workflow analysis and the development of intelligent computer-assisted surgical systems. However, accurate phase recognition remains challenging due to visual similarities between surgical phases and variability in surgical workflow duration. This work investigates multiple machine learning and deep learning approaches for cataract surgery phase recognition using the CAT-21 dataset, with a particular focus on incorporating temporal information. The evaluated approaches include Logistic Regression, Convolutional Neural Networks (CNNs), CNNs with temporal pooling, recurrent CNN+RNN architectures, and 3D Convolutional Neural Networks (3D CNNs). Temporal context was additionally incorporated through sliding window strategies to smooth prediction outputs and reduce abrupt temporal oscillations between consecutive predictions. Experimental results demonstrated that incorporating temporal information substantially improved classification performance and prediction stability across all evaluated architectures. Among the evaluated models, the CNN+RNN architecture achieved the strongest overall performance, obtaining a macro-average F1-score of 0.7653 on the CAT-21 testing set. Overall, the findings highlight the importance of temporal modelling for surgical workflow analysis and cataract surgery phase recognition.

Index Terms—Cataract surgery, surgical phase recognition, temporal modelling, CNN, RNN, 3D CNN.

CODE AVAILABILITY

The implementation and experimental pipeline used in this work are publicly available at:

https://github.com/stefanosPanteli/EPL445_CataractVideoPhaseClassification

I. INTRODUCTION

Cataract surgery is one of the most commonly performed ophthalmic procedures worldwide. Automatic recognition of surgical phases from video recordings can support workflow analysis, post-operative evaluation, and the development of intelligent computer-assisted surgical systems. More specifically, surgical phase recognition aims to classify each video frame according to the current stage of the operation.

In this work, we use the CAT-21 cataract surgery dataset, which contains annotated cataract surgery videos divided into predefined surgical phases. Each surgical procedure follows a structured temporal workflow, where phases appear sequentially throughout the operation. However, phase durations may vary considerably across surgeries, introducing additional challenges for accurate phase recognition.

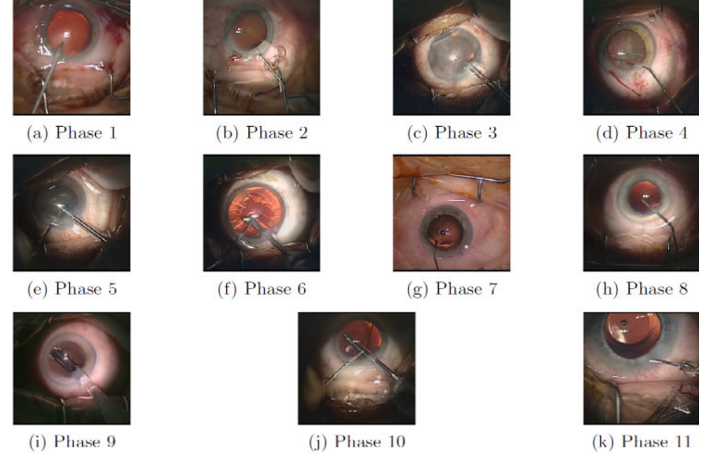


Fig. 1. Example frames corresponding to different cataract surgery phases in the CAT-21 dataset.

To better understand the temporal structure of the dataset, we analysed the annotated phase progression across surgical cases. As illustrated in Figure 2, cataract surgery procedures follow a relatively consistent sequential workflow despite differences in operation duration. This temporal continuity suggests that neighbouring frames contain useful contextual information for classification.

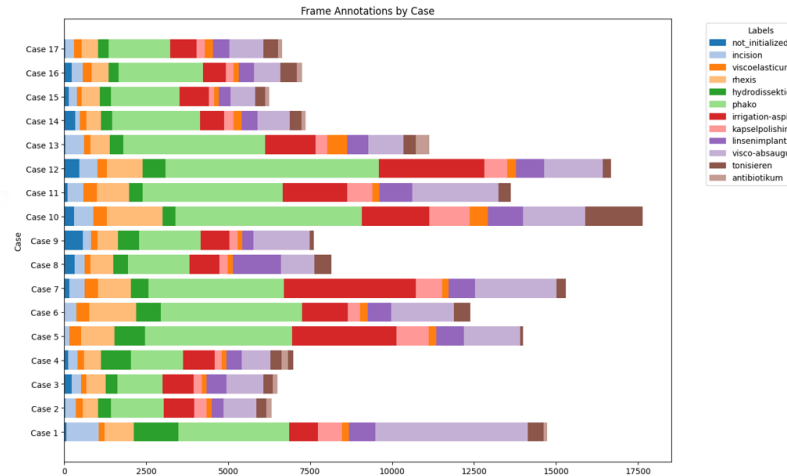


Fig. 2. Visualisation of annotated surgical phase progression across different cataract surgery cases.

In addition, several phases exhibit different duration distributions and variability patterns. Certain phases, such as *Phacoemulsification*, show substantially higher variability and longer-duration outliers, increasing the complexity of frame-level classification.

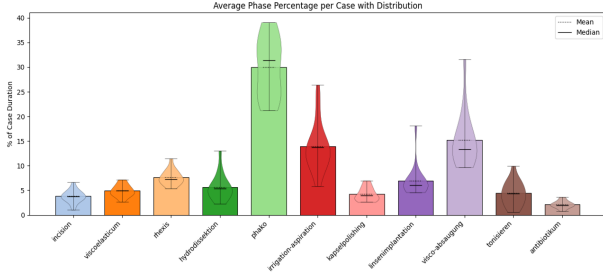


Fig. 3. Average duration and variability of cataract surgery phases across the CAT-21 dataset.

Although many surgical phases share similar visual characteristics, their temporal ordering provides important information for distinguishing between phases. Consequently, approaches that incorporate temporal context may better capture the evolution of surgical workflows compared to purely frame-based methods.

Temporal modelling has also been widely explored in surgical workflow analysis using recurrent and temporal convolutional architectures [1]–[3].

Previous work by Primus et al. [4] demonstrated that incorporating temporal information into convolutional neural network architectures can significantly improve cataract surgery phase recognition performance.

Motivated by the importance of temporal information, this project investigates multiple approaches for cataract surgery phase recognition, including logistic regression models using extracted metadata, convolutional neural networks, temporal pooling approaches, recurrent neural networks, and 3D convolutional architectures.

The goal of this work is to examine how incorporating temporal information affects surgical phase recognition performance and to compare the strengths and limitations of different machine learning and deep learning approaches for surgical workflow analysis.

II. METHODOLOGY

A. Introduction

This work investigates multiple machine learning and deep learning approaches for cataract surgery phase recognition, ranging from metadata-based baselines to architectures incorporating temporal information. Since cataract surgery follows a structured workflow, neighbouring frames often contain useful contextual information regarding the current surgical phase.

To reduce computational cost and redundancy between highly similar frames, videos were downsampled during pre-processing using different frame sampling intervals. In addition, sliding window approaches were employed to provide

short-term temporal context by grouping consecutive frames into windows of varying sizes.

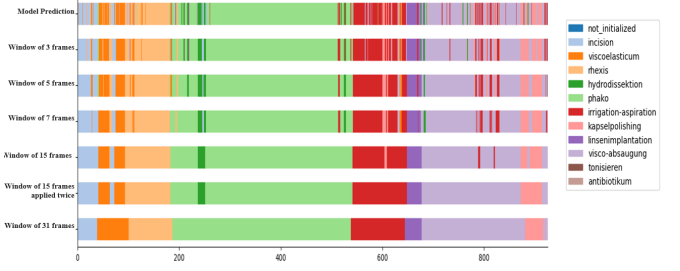


Fig. 4. Illustration of temporal sliding window configurations used during classification.

Several preprocessing techniques were additionally applied before training. Frames labelled as *not_initialized* were removed, videos were downsampled by sampling every 15 frames, all frames were resized to 224×224 pixels, and colour channels were converted from BGR to RGB format. Additional metadata features such as colour statistics, entropy, edge information, and temporal position were also extracted. To mitigate class imbalance, overrepresented phases were downsampled, while underrepresented phases were augmented using transformations such as rotation, brightness adjustment, contrast modification, and noise injection.

To further incorporate temporal progression information, a lightweight multilayer perceptron (MLP) was introduced using normalized frame position features. The extracted temporal embedding was combined with the visual representations learned by the classification models.

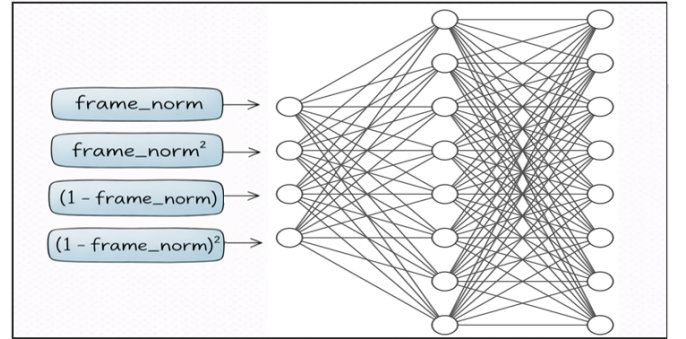


Fig. 5. FrameNorm multilayer perceptron used for temporal progression encoding.

B. Classification Models

Multiple machine learning and deep learning architectures were evaluated for cataract surgery phase recognition, including Logistic Regression, Convolutional Neural Networks (CNNs), CNNs with temporal pooling, CNN+RNN models, and 3D CNN architectures.

1) *Logistic Regression*: Logistic Regression was used as a metadata-based baseline classifier. The model received hand-crafted features extracted from video frames, including colour

statistics, entropy, edge information, and temporal descriptors derived from the FrameNorm representation.

Although simpler than deep learning approaches, Logistic Regression provided an interpretable and computationally efficient baseline for comparison against temporally-aware architectures.

2) *Convolutional Neural Network (CNN)*: A pretrained ResNet-18 architecture was employed for frame-level classification. Each input frame was resized to 224×224 pixels before being processed by the CNN feature extractor.

To incorporate temporal progression information, visual embeddings were combined with the FrameNorm temporal embedding before classification.

3) *CNN with Short Window Pooling*: To capture short-term temporal dependencies, consecutive frames were grouped into temporal windows before classification. Each frame was processed independently by the CNN backbone, while the extracted embeddings were merged using mean pooling to produce a unified temporal representation.

This approach improved temporal consistency compared to isolated frame classification.

4) *CNN with Recurrent Neural Network (CNN+RNN)*: A hybrid CNN+RNN architecture was also investigated for modelling longer-term temporal dependencies. A ResNet-18 backbone was first used for visual feature extraction, while a recurrent neural network based on GRUs modelled temporal transitions between surgical phases.

By jointly modelling visual appearance and temporal evolution, the CNN+RNN architecture achieved the strongest overall performance among the evaluated approaches.

5) *3D Convolutional Neural Network (3D CNN)*: A pretrained *r3d_18* architecture was additionally evaluated for spatiotemporal modelling. Unlike 2D CNNs, 3D CNNs jointly process spatial and temporal information directly from short video clips.

It also exhibited signs of overfitting due to the limited size of the CAT-21 dataset.

TABLE I
BEST-PERFORMING CONFIGURATION FOR EACH EVALUATED CLASSIFICATION ARCHITECTURE.

Model	Cfg.	Acc.	Prec.	Rec.	F1
LR	win31	0.5632	0.5497	0.5632	0.3531
CNN	unbalanced_win15	0.7546	0.7674	0.7546	0.5090
CNN+Pool	balanced_win7	0.8714	0.8857	0.8714	0.7159
CNN+RNN	balanced_win15	0.8485	0.8759	0.8485	0.7198
3D CNN	unbalanced_win7	0.6995	0.8482	0.6995	0.5579

C. Best Performing Temporal Architectures

The experimental results demonstrated that incorporating temporal information substantially improved cataract surgery phase recognition performance. Among the evaluated approaches, the CNN temporal pooling, CNN+RNN, and 3D CNN architectures achieved the strongest overall results.

Among all evaluated models, the CNN+RNN architecture achieved the best overall performance by effectively mod-

elling long-term temporal dependencies throughout the surgical workflow.

TABLE II
ARCHITECTURE OF THE BEST-PERFORMING CNN+RNN MODEL.

Layer	Type	Output	Configuration
Input	Video clip	$31 \times 3 \times 224 \times 224$	RGB frames
CNN Backbone	ResNet18	512	Pretrained
Temporal Modeling	RNN	128	3 layers
Fully Connected	Linear	128	ReLU
Output	Classification	10 classes	Softmax

D. Extended Training Experiments

Following the initial experiments, additional evaluations were conducted using all available CAT-21 training videos to further investigate the strongest temporal architectures.

For these experiments, frames were sampled every 5 frames to preserve denser temporal information while reducing redundancy. Based on the preliminary results, the CNN temporal pooling, CNN+RNN, and 3D CNN architectures were selected for further experimentation.

III. RESULTS

A. Comparison of Best Extended Configurations

Table III summarizes the best-performing configuration obtained from each temporal architecture during the extended training experiments.

TABLE III
BEST-PERFORMING TEMPORAL ARCHITECTURES UNDER THE EXTENDED TRAINING SETUP.

Architecture	Cfg.	Acc.	Prec.	Rec.	Macro F1	Weighted F1
CNN+Pool	win31	0.8044	0.8237	0.8044	0.7345	0.7996
CNN+RNN	win31	0.8154	0.8285	0.8154	0.7653	0.8112
3D CNN	win31	0.7134	0.7840	0.7134	0.6222	0.7185

Among the evaluated approaches, the CNN+RNN architecture achieved the strongest overall performance, obtaining the highest macro-average F1-score of 0.7653. The CNN temporal pooling model also achieved strong performance, while the 3D CNN architecture showed lower overall performance and signs of overfitting.

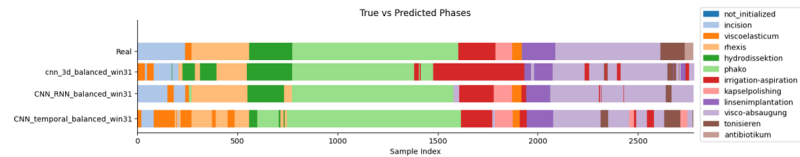


Fig. 6. Comparison of prediction consistency across the evaluated temporal architectures on Case 18, including the 3D CNN, CNN+RNN, and CNN temporal pooling models.

Figures 6 illustrate the temporal prediction behaviour of the evaluated architectures on Case 18 using the best-performing sliding window configuration. Increasing the sliding window

size generally improved prediction smoothness and reduced isolated phase fluctuations.

The CNN+RNN architecture produced the smoothest temporal transitions and the strongest alignment with the ground-truth workflow structure. In comparison, the 3D CNN model remained noisier and exhibited signs of overfitting.

B. Best Performing Model Analysis

Among all evaluated approaches, the CNN+RNN architecture achieved the strongest overall performance. The best-performing configuration used a temporal window size of 31 frames under the balanced training setup, obtaining a macro-average F1-score of 0.7653 on the CAT-21 testing set.

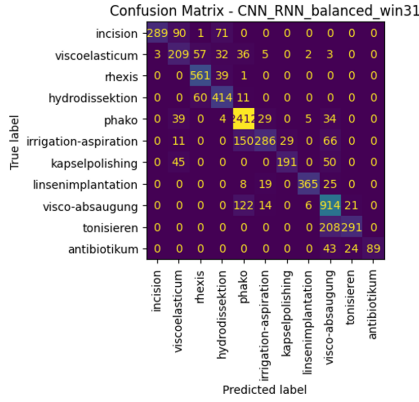


Fig. 7. Confusion matrix of the best-performing CNN+RNN configuration.

Figure 7 shows that the model successfully captured the overall sequential workflow structure of cataract surgery. Most predictions were concentrated around the diagonal of the confusion matrix, indicating strong classification performance across the majority of surgical phases.

Most classification errors occurred between temporally neighbouring phases, suggesting that the model learned the general surgical progression but occasionally struggled during short transition periods.

IV. DISCUSSION

This work investigated multiple machine learning and deep learning approaches for cataract surgery phase recognition using the CAT-21 dataset. Overall, the results demonstrated that incorporating temporal information improved classification performance and prediction stability across all evaluated architectures.

Among the evaluated approaches, the CNN+RNN architecture achieved the strongest overall performance, obtaining a macro-average F1-score of 0.7653 on the CAT-21 testing set. The CNN temporal pooling architecture also achieved strong performance, while the 3D CNN architecture showed lower overall performance and signs of overfitting.

The obtained results are consistent with previous work demonstrating the importance of temporal modelling for surgical workflow analysis and phase recognition [1], [3], [4]. Several limitations should be considered. First, the CAT-21 dataset

contains a relatively limited number of annotated surgical procedures, restricting workflow diversity during training. In addition, some architectures, particularly the 3D CNN models, exhibited signs of overfitting.

Future work could further improve performance through the use of larger datasets such as the Cataract-101 Video Dataset and more advanced temporal architectures, including temporal convolutional and transformer-based approaches [3].

V. CONCLUSION

This work investigated multiple machine learning and deep learning approaches for cataract surgery phase recognition using the CAT-21 dataset. The experimental results demonstrated that incorporating temporal information substantially improved classification performance and prediction stability.

Among the evaluated approaches, the CNN+RNN architecture achieved the strongest overall performance, obtaining a macro-average F1-score of 0.7653 on the CAT-21 testing set.

Overall, the findings highlight the importance of temporal modelling for surgical workflow analysis. Future work could further improve performance using larger datasets and more advanced temporal deep learning architectures.

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